```
clc; clear;
Wp=0.4; Ws=0.55; Rp=1; As=40;

[N,Wn]=cheb1ord(Wp,Ws,Rp,As);
[b,a]=cheby1(N,Rp,Wn);

fprintf('Filter Order = %d\n',N);
```

```
Filter Order = 6
```

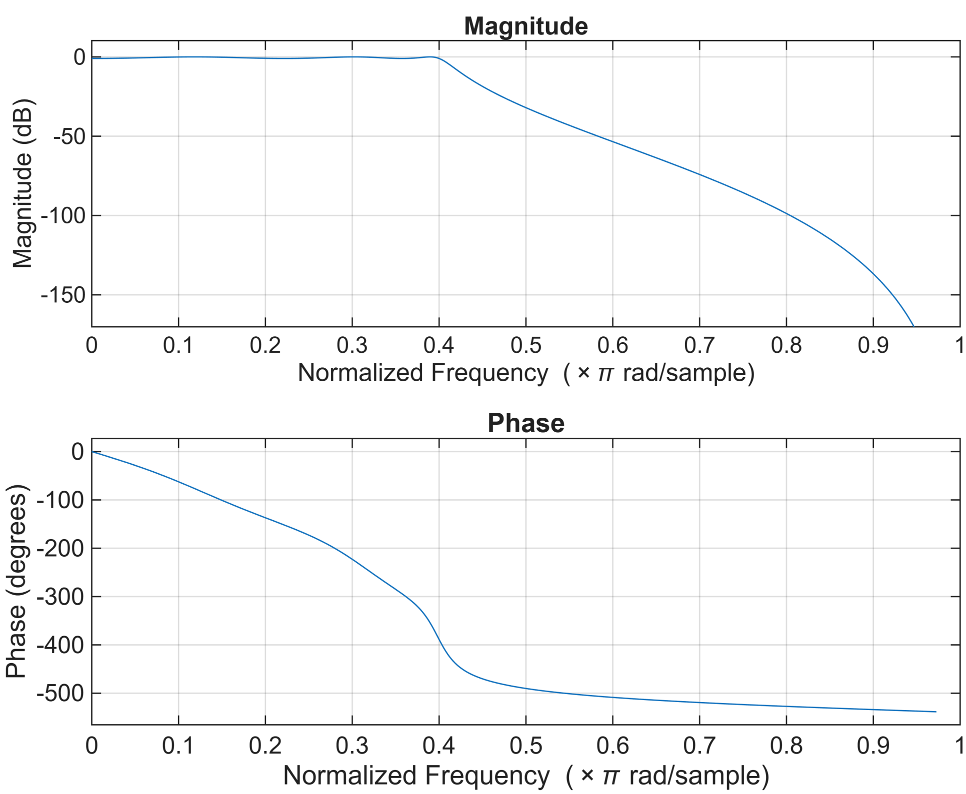
```
fprintf('Cutoff Frequency = %.2f * pi rad/sample\n',Wn);
```

```
Cutoff Frequency = 0.40 * pi rad/sample
```

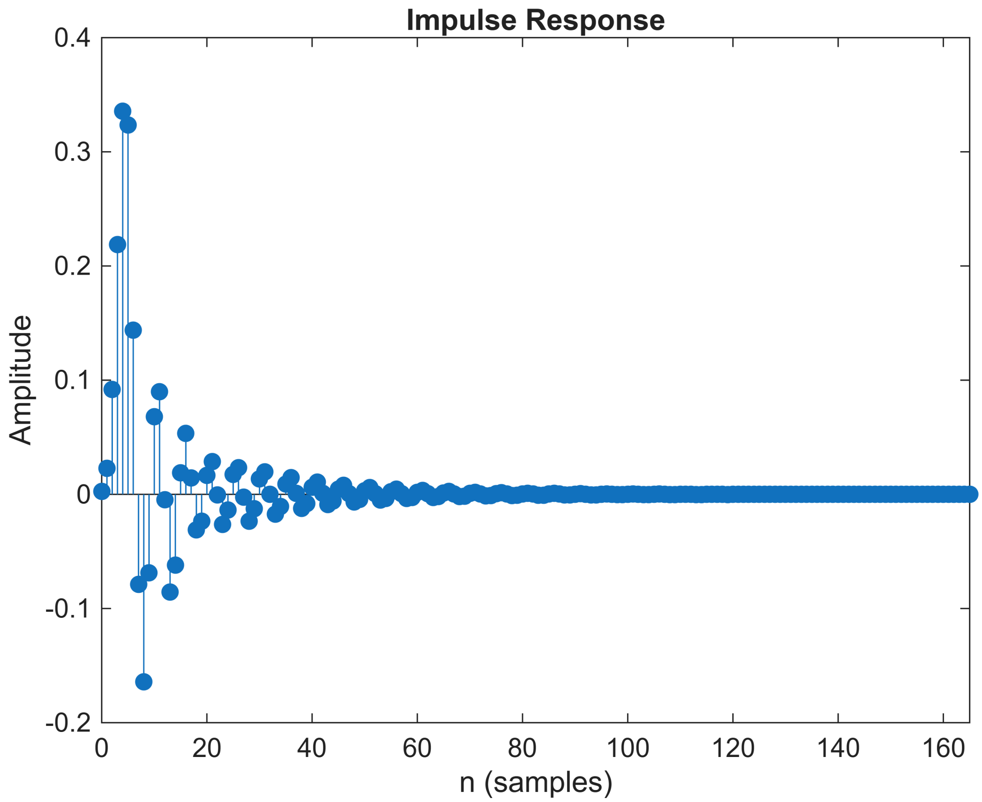
```
if all(abs(roots(a))<1)
    disp('System is STABLE');
else
    disp('System is UNSTABLE');
end
```

```
System is STABLE
```

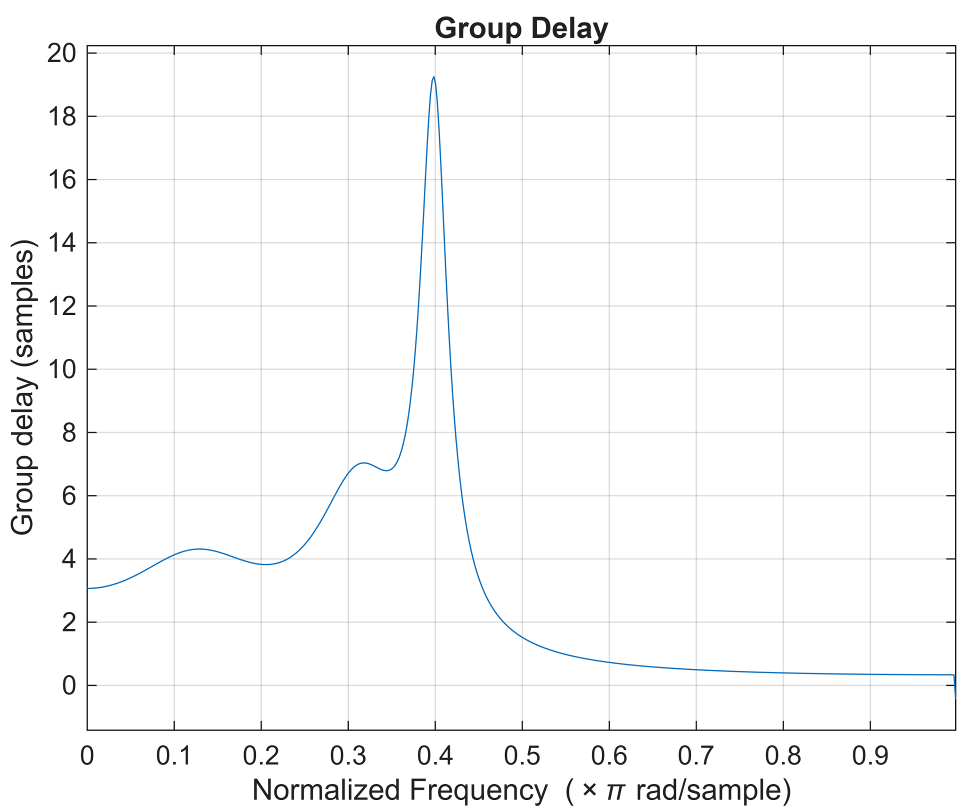
```
% -------- GRAPHS --------
figure; freqz(b,a);     % Magnitude & Phase
```




```
figure; impz(b,a);      % Impulse Response
```




```
figure; grpdelay(b,a);  % Group Delay
```



```
figure; zplane(b,a);    % Pole-Zero
```